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Small Seeds, Thin Markets: Climate Risk, Supply Chains, and the Economics of Sesame from Farm to Fast Food

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Abstract. We trace the full causal chain from climate variability to the economics of the hamburger bun, using the global sesame seed market as the case. World production was 6.68 million tonnes in 2024 (Observed, A), concentrated in a handful of rainfed smallholder systems across Africa and Asia, with roughly one third entering international trade. Two null results anchor the findings. First, at country scale over 2000–2024, growing-season rainfall and temperature anomalies explain essentially none of the year-to-year variation in detrended sesame yields (pooled $r = +0.02$, $p = 0.77$, $n = 200$; Reconstructed, B): severe weather moves yields in documented episodes, but national aggregates dilute the signal, and conflict has dominated the recent collapses. Second, annual world production changes explain none of annual US import price changes ($R^2 = 0.00$, $n = 18$; Reconstructed, B): origin substitution absorbs most shocks, as the 2023–24 Sudan episode demonstrates (-29% origin output, flat world price). Where shocks do transmit, the price elasticity of demand is about -0.65 (Modelled, B), estimated from India's 2022 collapse. Downstream, the price channel is economically irrelevant: a 50% sesame price spike costs the average US McDonald's restaurant about \$98 per year, 0.02% of EBITDA (Modelled, B). The binding vulnerability is operational. Allergen law has locked sesame into bun formulas, supplier re-approval takes 2 to 6 weeks, and three Class I allergen recalls since 2022 show the failure mode. Five scenarios for 2027–2035 put the modal outcome at orderly growth (55%), with a 12% tail of severe multi-region failure (+31 to +46% prices) and a 10% risk of sustained structural deficit (+50 to +100%).

Keywords: sesame, supply chains, climate risk, commodity prices, food service, price elasticity
JEL codes: Q11, Q54, L66, D43

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Contents

1 Introduction **3**

2 Institutional background **3**

3 Related literature **4**

4 Data **5**

4.1 Sources and definitions 5

4.2 Summary statistics 6

5 Methodology **6**

6 Results **7**

6.1 The supply chain: concentrated, rainfed, and already shocked 7

6.2 Climate and yields: the null at country scale, the signal in episodes 9

6.3 Prices: substitution absorbs most shocks 11

6.4 Downstream: the price channel is irrelevant; the operational channel binds 12

7 Scenarios, 2027–2035 **14**

8 Limitations, robustness, and falsifiability **16**

9 Conclusion **17**

A Data appendix **20**

1 Introduction

The sesame seed on a hamburger bun weighs about 1.68 grams. The global market that produces it weighs about 6.7 million tonnes a year, grown mostly by smallholders on rainfed plots across Sudan, India, Myanmar, Ethiopia, and a dozen other countries, then cleaned, hulled, shipped, baked, and served through one of the most rigid procurement systems in food service. This paper asks a simple question with a long causal chain: how does climate variability at the farm end up affecting the economics of the restaurant, and where in that chain do small disruptions create disproportionate consequences?

The question matters now for three reasons. First, the world's largest sesame producer, Sudan, lost 59% of its output between 2020 and 2024, and the cause was war, not weather, which raises the question of what a climate-driven shock of similar size would do. Second, the United States made sesame a major food allergen in 2023, and the industry's response, adding sesame to bun formulas rather than segregating it, reduced the system's flexibility exactly when climate exposure is rising. Third, the fast growers in sesame production, Burkina Faso, Mozambique, Ethiopia, Brazil, are expanding at 14 to 16% a year from small bases, and nobody has checked whether their rainfall risks are correlated with the incumbents they are replacing.

We assemble four linked bodies of evidence: a supply-chain map from FAOSTAT production and UN Comtrade trade data; a yield-weather panel for eight producing countries over 2000–2024 matched to satellite-era weather; a price and shock-episode analysis with elasticity estimation; and a downstream cost model for three quick-service chains. The headline results are two nulls and one asymmetry. The nulls: country-level weather does not explain country-level yields, and world production does not explain world prices, because substitution and aggregation absorb most shocks. The asymmetry: when shocks do reach the restaurant, they arrive through operations, not prices. A 50% sesame price spike costs a McDonald's franchisee about $98 a year. A two-week shortage of approved food-grade sesame forces spec changes, re-labeling, and re-approval across the distribution system.

The rest of the paper proceeds as follows. Section 2 describes how the sesame market actually works, from smallholder plots to approved bakeries. Section 3 reviews the related literature. Section 4 documents the data. Section 5 sets out the methodology. Section 6 presents the results in four parts: supply chain, climate and yields, prices, and downstream economics. Section 7 builds the 2027–2035 scenarios. Section 8 discusses limitations, robustness, and falsifiability. Section 9 concludes.

2 Institutional background

Sesame (*Sesamum indicum* L.) is a warm-season annual oilseed grown overwhelmingly by smallholders on rainfed land. In Sudan, about 60% of output comes from traditional rainfed farms (ETI Sweden, 2023) of 2 to 50 hectares and 40% from semi-mechanized rainfed farms (Observed, B; ETI Sweden 2023). In India, Myanmar, Nigeria, and Tanzania, 80 to 90% of volume comes from farms

under 5 hectares (Observed, B). There is essentially no irrigation, no contract farming outside narrow high-end export niches, and no traceability.

The chain runs: farmer to village collector to regional auction to exporter, who cleans, sorts, bags, and containerizes raw seed. Processing into hulled sesame, tahini, or oil happens at the destination or at specialized hulling plants. The industry is fragmented: India's processing sits in dozens of firms around Unjha, Gujarat; Greece's Haitoglou processes up to 150 tonnes a day; Ethiopia's Selet Hulling runs 10,000 tonnes a year with its own 300-hectare organic farm, one of the few documented cases of processor farm ownership (Observed, B).

About 36% of world output enters international trade (Reconstructed, B). China absorbs roughly half of traded sesame, about 913,000 tonnes in 2023 (Observed, A), for crushing into oil. Japan, Korea, Turkey, and the Middle East take most of the rest. The United States imports only about 32,000 tonnes, mostly from India. Trade is thin and concentrated: the top bilateral corridor, Niger to China, moved 193,000 tonnes in 2023 (Observed, A).

At the restaurant end, procurement is rigid. McDonald's US franchisees must buy buns from approved bakeries, Fresh Start (Aryzta), Aspire, C.H. Guenther, Klosterman, Conagra, Bimbo (Observed, B). There is no sesame futures market with usable liquidity; the NCDEX sesame contract exists but trades too thinly to hedge (Observed, B). Bakeries forward-buy seed through suppliers. Commodity cost changes pass through to franchisees at contract rollover.

The FASTER Act, effective January 1, 2023, declared sesame a major food allergen in the United States. Rather than segregate sesame, major bakers added it to formulas and declared it: Wendy's, Chick-fil-A, Olive Garden, Pepperidge Farm, Bimbo, Martin's, Flowers Foods (Observed, B). This raised structural sesame demand and removed the plain-bun fallback for those products. In-N-Out reversed course in September 2026, removing sesame flour from its buns (Observed, C).

3 Related literature

The agronomy literature establishes sesame's asymmetric climate sensitivity. The crop tolerates drought through a deep taproot, but the seedling stage is moisture-sensitive and the flowering-to-capsule-fill window is critical (infonet-biovision; Kadam et al. 2020). Waterlogging is the sharper risk: in a 119-genotype screen, all plants died under 72 hours of waterlogging (MDPI Plants, 2021), and poor drainage causes 44% yield losses, up to 50–90% in severe cases (Wei et al., 2013). Heat above 40°C inhibits pollination (Kadam et al., 2020). The drought-linked pathogen charcoal rot (*Macrophomina phaseolina*) causes 25–100% losses and is favored by hot, dry conditions (Vyas 1981; Rajendra Prasad et al. 2022).

On ENSO teleconnections to sesame regions, the climatology is standard: El Niño weakens the Indian monsoon and the Sahelian rains; La Niña strengthens them (NOAA CPC, 2026), with East Africa split by season. In Ethiopia, El Niño is associated with June–September drought and La Niña with excess rain and flooding (Degefu et al., 2017; Korecha and Barnston, 2007). Sesame-

specific econometric work linking these teleconnections to yields at scale is sparse, which is the gap this paper's panel addresses.

The commodity-economics literature on thin agricultural markets emphasizes substitution and inventory buffers (see, e.g., the food-price spike literature of 2007–08). Our finding that origin shocks need not move world prices parallels the broader result that trade networks absorb localized supply failures when alternate origins hold surplus.

On the downstream side, the food-service procurement literature documents approved-supplier rigidity and the cost of specification changes, but we found no published quantification of sesame cost shares at the restaurant level. The allergen-recall literature documents the Class I recall mechanism; three sesame recalls since 2022 are a new data point.

This paper's contribution is the linked chain: farm weather to yields to prices to restaurant economics, with each link estimated separately and the nulls reported rather than buried.

4 Data

4.1 Sources and definitions

Table 1 inventories the datasets. Production comes from FAOSTAT QCL (item code 289, sesame seed) via the ElkassabgiData mirror, because the FAOSTAT direct API was unreachable (Cloudflare 521) during retrieval on 2026-10-07/08; a spot check against a published FAOSTAT-derived figure (Tanzania 2024) matched exactly (Observed, A). Trade comes from the UN Comtrade public preview API, HS 120740 (Observed, A). India and Sudan do not report to Comtrade; their exports appear only in partners' mirror import data, so true world exports exceed reported exports by roughly 350–400 kt in 2023 (Reconstructed, B).

Weather comes from NASA POWER (precipitation, 2-meter temperature) for seven countries and ERA5 via Open-Meteo for Myanmar, after cross-validation rejected ERA5 for Sudan and Ethiopia (factor-six precipitation discrepancies against POWER and against the observed record harvest) and rejected POWER for Myanmar (a spurious pre-2008 precipitation break). An early Myanmar correlation computed with POWER data vanished under ERA5 and is documented as a reanalysis artifact. ENSO phase uses NOAA CPC monthly ONI averaged over each country's growing season.

Prices are US import unit values (CIF, HS 1207.40), Ethiopia export unit values (FOB), and China import unit values (CIF), all from Comtrade (Observed, A). Farmgate prices come from FAOSTAT producer prices via the mirror (Observed, B).

Downstream inputs come from company filings (McDonald's 2025 10-K; RBI and Wendy's 2025 results), QSR 50 2026 unit counts, a disclosed bun specification sheet (Låntmannen Unibake, January 2026: 2% sesame on an 84 g bun), and trade-press sesame price benchmarks (Observed, A/B as tagged in text).

Table 1: Data inventory

Dataset	Source	Coverage
Production by country	FAOSTAT QCL via mirror	2000–2024, 4,540 rows
Trade flows	UN Comtrade HS 120740	2015–2024
Bilateral imports	UN Comtrade HS 120740	2022–2024, 12 importers
Yield-weather panel	FAOSTAT + NASA POWER/ERA5 + NOAA	8 countries × 25 yrs
Price series	UN Comtrade HS 1207.40	US/Ethiopia/China
Farmgate prices	FAOSTAT PP via mirror	Ethiopia, Myanmar, China, Mexico
Chain financials	10-Ks, QSR 50	2025

Notes: FAOSTAT direct API unreachable during retrieval; mirror cross-checked. India and Sudan absent from Comtrade reporter data.
Source: Author computations from sources listed.

4.2 Summary statistics

Table 2 summarizes the estimation samples. World production averaged 5.0 million tonnes over 2000–2024 with a standard deviation of 1.2 million; the US sesame import price averaged \$1.76/kg with a standard deviation of \$0.63.

Table 2: Summary statistics, estimation samples

Series	Mean	Std. dev.	Min	Max	N
World production (Mt)	5.02	1.21	2.86	7.02	25
US import price, CIF (\$/kg)	1.76	0.63	0.81	3.04	24
Detrended yield anomaly (%)	0.0	14.8	−45	172	200
Growing-season rain anomaly (%)	0.0	28.4	−52	215	200
McDonald's US sesame (t/yr)	1,681	–	–	–	1 (est.)

Notes: Yield and rain anomalies are percent deviations from country OLS trend (yields) and 1991–2020 climatology (rain). The 172% yield anomaly is Nigeria 2012, a FAOSTAT reporting artifact. McDonald's tonnage is a single modeled estimate, not a sample.
Source: Author computations; see Appendix A.

5 Methodology

The yield-weather panel covers eight countries (Sudan, Myanmar, India, Nigeria, Tanzania, Ethiopia, China, Mexico) over 2000–2024. For each country-year we compute the detrended yield anomaly (percent deviation from an OLS linear trend; Sudan split at the 2011 South Sudan partition) and the growing-season weather anomaly (precipitation summed and temperature averaged over 2–5 sesame-belt points, vs 1991–2020 climatology). We report Pearson correlations per country and pooled, plus Welch t-tests of yield anomalies by ONI phase. With $n = 25$ per country, the design has roughly 80% power to detect $|r| \geq 0.52$ at $p < 0.05$.

Price elasticity is estimated by arc elasticity from the cleanest supply shock in the record, India's

2022 collapse (India −50.4%, world −11.7%, US import price +22.3%), giving a central demand elasticity of −0.65 with a range of −0.50 to −1.00 (Modelled, B). The 2022 estimate is an upper-magnitude bound because concurrent broad commodity inflation may inflate the price response. Scenario price impacts use $\%\Delta P = \%\Delta Q/|\varepsilon_d|$. An upper-bound variant multiplies the shock by the export share (only ∼36% of output is traded), since a global output cut concentrates on exportable surplus.

Downstream costs use 1.68 g sesame per seeded bun (Observed, B; one disclosed European spec sheet) and a $1.60/kg food-grade benchmark (2024–2026 trade press; Assumed, C for contract prices, which no chain discloses). Chain tonnages scale units × modeled burgers per day × seeded-bun share × 365 (Modelled, C).

Evidence grading follows the paper standard: every reported number is tagged Observed, Reconstructed, Modelled, or Assumed, with source quality A (official/primary), B (reputable secondary), or C (provisional/estimated). This is a retrospective analysis; there was no pre-registered plan.

6 Results

6.1 The supply chain: concentrated, rainfed, and already shocked

World sesame production was 6.68 million tonnes in 2024 (FAOSTAT, 2026) (Observed, A). The top ten producers are shown in Figure 1. On a five-year average basis Sudan is the largest producer at 930,000 tonnes, but its 2024 output of 626,000 tonnes was 59% below its 2020 peak of 1.53 million tonnes (Observed, A), a collapse caused by the April 2023 war (Chatham House, 2024). India was the largest single-year producer in 2024 at 866,000 tonnes. China is the only top producer in structural decline (−3.1%/yr CAGR). The fastest structural growers are Burkina Faso (+16.2%/yr), Mozambique (+15.9%/yr), Ethiopia (+14.9%/yr), and Brazil (+14.7%/yr) (all Reconstructed, B).

Figure 2 shows production paths since 2000. The visual story is Sudan's wartime collapse against steady growth elsewhere, and China's long decline as it shifted from producer to the world's dominant importer.

Trade is thin and China-centric. Reported world exports ran 1.13 to 1.58 million tonnes a year over 2015–2024; true exports are near 1.9 million tonnes once the India-Sudan mirror gap is added (Reconstructed, B). China took 913,000 tonnes in 2023, about 49% of reported imports (UN Comtrade, 2026) (Observed, A). The largest bilateral corridors are shown in Figure 3: Niger to China (193 kt), Pakistan to China (167 kt), Tanzania to China (130 kt). The United States imports only about 32,000 tonnes, 57% from India (Observed, A).

Farm structure is uniformly smallholder and rainfed: 80–90% of volume from farms under 5 hectares in India, Myanmar, Nigeria, and Tanzania; Sudan splits 60/40 between traditional smallholders and semi-mechanized rainfed farms (MDPI Agronomy, 2021) (all Observed, B). There is no irrigation, no contract farming outside narrow export niches, and no traceability. Inventories are purely commercial: Ethiopia's ECX warehouse receipts expire after 60 days with penalties designed to

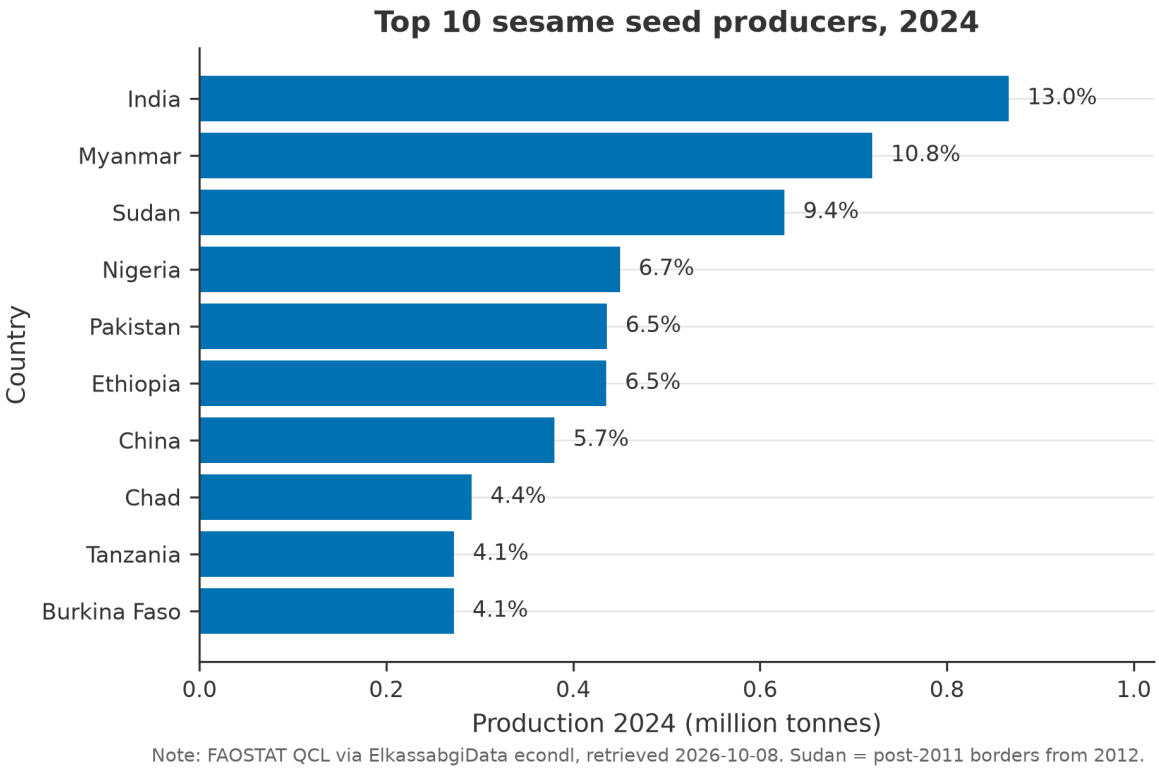


Figure 1: Top 10 sesame seed producers, 2024
Notes: Shares of 6.68 Mt world total. Sudan reflects post-2011 borders from 2012.
Source: FAOSTAT QCL via ElkassabgiData, retrieved 2026-10-08.

discourage stockpiling (USDA FAS, 2021), and Qingdao port held about 379,000 tonnes in September 2026, roughly four months of Chinese imports (Commodity Board, 2026) (Observed, B). No government strategic sesame reserve was documented anywhere.

On processors: the brief's central case study does not survive verification. DIPASA (Mexico) is a verified sesame processor and exporter with sister companies in the US and Netherlands, and a 2021 seed-company announcement described its clients as including "familiar fast food and global food brands" without naming any (Seed World, 2021) (Observed, B). No public record was found of DIPASA's ownership, tonnage, farm ownership, or any commercial relationship with McDonald's, Burger King, or Wendy's. The only verified processor-to-industrial-bakery link found anywhere in the research is Sesajal's ISO 22000 certification undertaken at Bimbo's request (Observed, B). Other documented processors include Selet Hulling (Ethiopia, 10,000 t/yr, with a 300-hectare organic farm, one of the few verified processor farm holdings), Haitoglou (Greece, up to 150 t/day), and Olam Agri (Turkey, Nigeria). India's hulling is fragmented across dozens of firms in Unjha, Gujarat, with no documented dominant share.

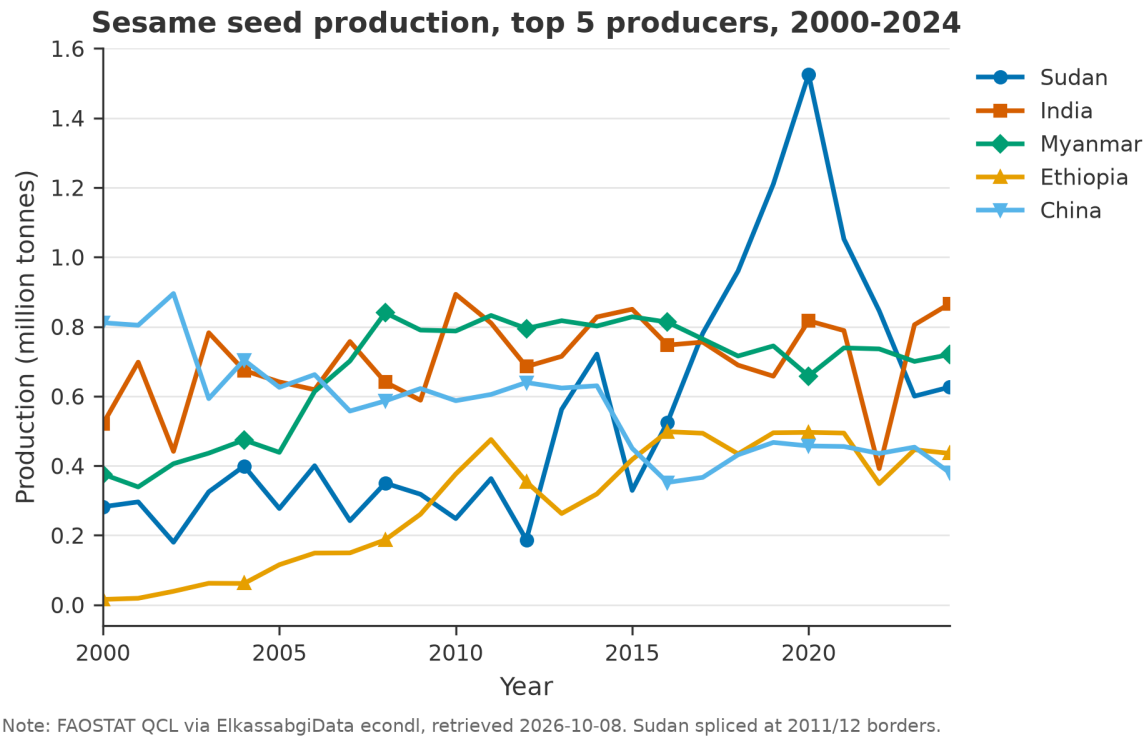


Figure 2: Sesame production, top 5 producers, 2000–2024 (million tonnes)
Notes: Sudan splices pre-2011 borders. FAOSTAT figures for Sudan 2022–2024 and Myanmar 2021–2024 are likely imputed under conflict conditions; treat with caution.
Source: FAOSTAT QCL via ElkassabgiData, retrieved 2026-10-08.

6.2 Climate and yields: the null at country scale, the signal in episodes

Sesame's agronomy sets up an asymmetric climate exposure. The crop tolerates drought via a deep taproot, but the seedling stage is moisture-sensitive, the flowering-to-capsule-fill window is critical, pollination is inhibited above 40°C, and waterlogging is lethal: all plants died under 72 hours of waterlogging in a 119-genotype screen (Observed, A; MDPI Plants 2021). The drought-linked pathogen charcoal rot causes 25–100% losses and is favored by hot, dry conditions (Vyas, 1981; Rajendra Prasad et al., 2022) (Observed, A).

Against this biology, the country-scale statistics are a null. Table 3 reports correlations between detrended yield anomalies and growing-season weather anomalies, 2000–2024. No country shows a significant linear relationship for rainfall or temperature; the pooled correlation is $r = +0.02$ ($p = 0.77$, $n = 200$).

Figure 4 shows the null visually: eight flat clouds. This is a finding, not a non-result. The design had 80% power to detect $|r| \geq 0.52$; the observed correlations are far below that. The interpretation is that national aggregates dilute sub-national weather (the documented mechanism operates at field and district scale), area adjustments buffer yield shocks, sesame's drought tolerance compresses the response in normal years, and reporting noise and conflict dominate several series.

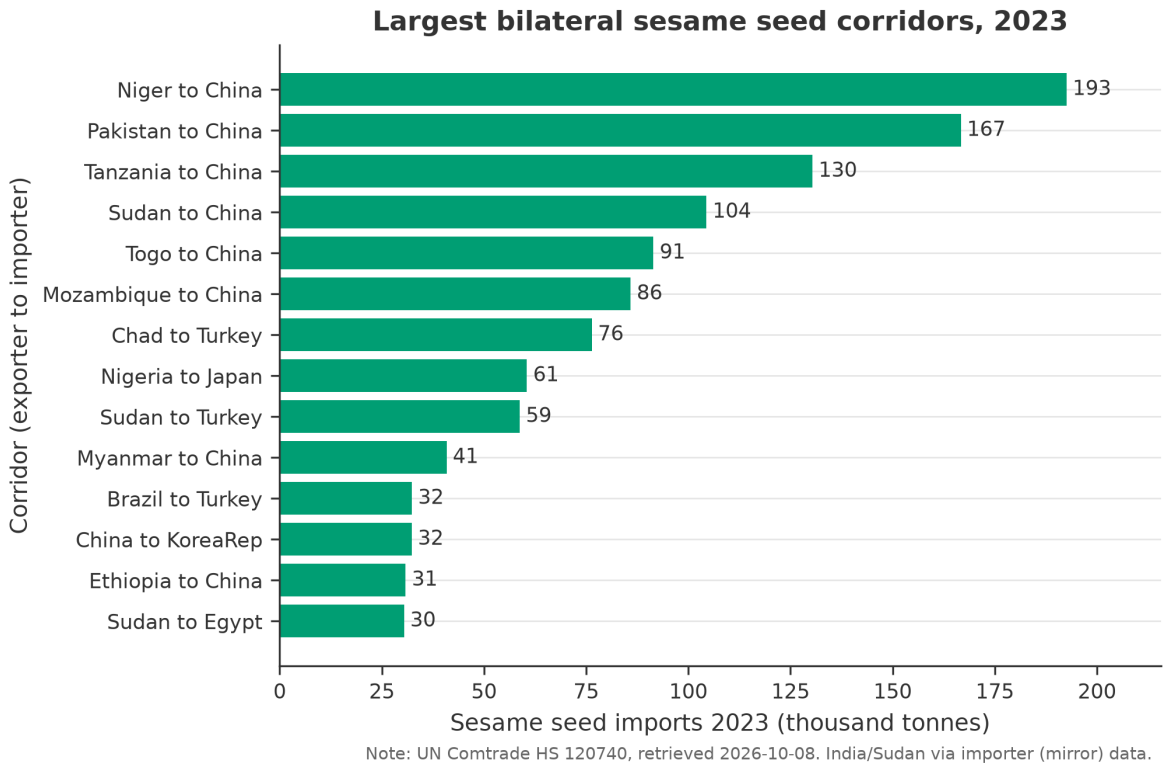


Figure 3: Largest bilateral sesame trade corridors, 2023 (thousand tonnes)
Notes: Importer-reported (mirror) data; India and Sudan do not report to Comtrade.
Source: UN Comtrade HS 120740, retrieved 2026-10-08.

ENSO phase shows one nominally significant result. Pooled across countries, La Niña years average -5.1% yield anomalies against $+0.4\%$ in neutral years (Welch $t = -2.18$, $p = 0.032$; Reconstructed, B), driven by Tanzania, Nigeria (data-quality caveat), and Mexico (Figure 5). El Niño averages -2.7%, not significant. Per-country ENSO samples are $n = 2$–4 and have almost no power; directionally, India and Sudan sit below their neutral means in El Niño years, consistent with the known monsoon and Sahel drying teleconnections, but the samples cannot distinguish this from noise.

The event studies show weather does move yields in severe episodes, which is why the null needs the episode-level complement. India's 2002 monsoon failure (El Niño) cut sesame yields 14% vs trend; 2009 cut 24% (Reconstructed, B). But 2015, another El Niño drought year, left Indian sesame near trend, a counterpoint with no established mechanism. Cyclone Nargis (2008) is a negative case: it hit Myanmar's rice delta, not the dry-zone sesame belt (MDPI Sustainability, 2020), and sesame yields rose 34%. Ethiopia's 2020–22 sesame collapse happened while the sesame belt received 39–60% above-normal rain: area halved under conflict while yields stayed within 6% of trend, so the collapse was conflict-driven by elimination (Observed, B). Sudan's 2023 output fell 29% while belt rain ran 83% above normal: not weather-driven (Observed, B). Figure 6 annotates the major episodes.

On correlated geographic risk: a single La Niña event can suppress yields across East Africa and

Table 3: Detrended yield anomaly vs growing-season weather anomaly, 2000–2024

Country	r (rain)	p	r (temp)	p
Sudan	−0.01	0.98	−0.02	0.93
Myanmar	−0.05	0.82	−0.21	0.31
India	+0.02	0.92	−0.11	0.60
Nigeria	+0.12	0.58	−0.06	0.78
Tanzania	+0.06	0.77	−0.07	0.74
Ethiopia	−0.09	0.68	−0.00	0.98
China	−0.24	0.24	+0.26	0.21
Mexico	+0.08	0.69	−0.02	0.92
Pooled ($n = 200$)	+0.02	0.77	−0.04	0.60

Notes: Pearson, two-tailed, $n = 25$ per country. Nigeria's series has reporting breaks; pooled results excluding Nigeria are likewise null. Weather: NASA POWER except Myanmar (ERA5). Source: Author computations from FAOSTAT, NASA POWER, ERA5, NOAA CPC.

South Asia simultaneously (the −5.1% pooled drag spans Tanzania, Nigeria, Mexico, with India and Sudan directionally negative in El Niño). The risk is real but modest in normal years; the historical record's worst global shock, India's 2022 −50% (cause still not established), cut world output only 11.7% because other origins held.

6.3 Prices: substitution absorbs most shocks

Figure 7 shows US sesame import unit values, 2000–2024. The series peak was $3.04/kg in 2014; the 2024 value was $2.73/kg (UN Comtrade, 2026) (Observed, A). The largest annual move, +74.8% in 2008, coincided with world production *rising* 2.6%: it was cross-commodity contagion from the food-and-energy boom, not a sesame supply shock.

The central price result is the second null: annual world production changes explain none of annual US import price changes (slope −0.06, s.e. 0.59, $R^2 = 0.00$, $n = 18$; Reconstructed, B), shown in Figure 8. Five documented shock episodes explain why. In 2022, India's −50.4% collapse (world −11.7%) lifted US prices +22.3%: the cleanest supply shock, too large to substitute quickly. In 2023–24, Sudan's −29% conflict-driven output loss left world prices flat (US −0.5%, China −5.7%) because Pakistan, Niger, and others replaced Sudan in China's buying (Observed, A/B). An origin shock of roughly 18% of world exports need not move world prices at all when alternate origins hold spare exportable surplus.

Elasticity comes from the 2022 India episode: arc elasticity −0.65, range −0.50 to −1.00 (Modelled, B), an upper-magnitude bound given concurrent broad commodity inflation. Scenario price impacts follow $\%\Delta P = \%\Delta Q/|\varepsilon_d|$: a 5% global supply cut lifts prices about 7.7%; 10% about 15.4%; 20% about 30.8%; 30% about 46.2%; 50% about 76.9% (Modelled, B), with the fan in Figure 9. An upper-bound variant notes that only ∼36% of output is traded, so a 10% output cut can remove ∼28% of exportable supply, implying up to +43% (Modelled, C).

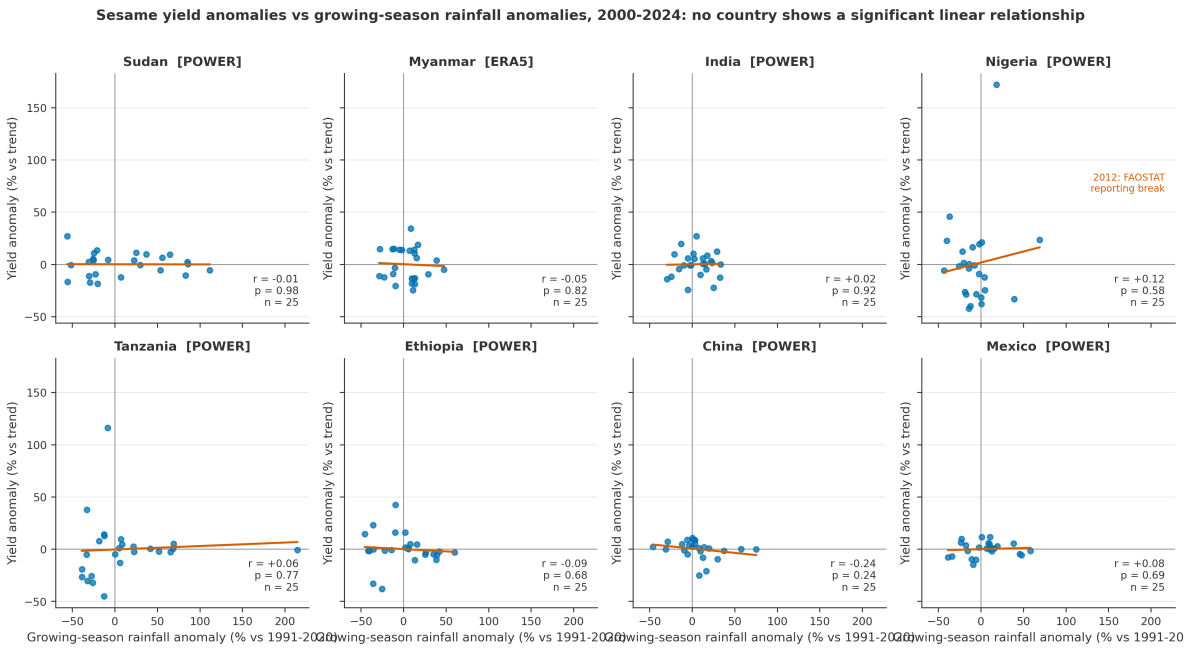


Figure 4: Yield anomalies vs growing-season rainfall anomalies, 2000–2024: no country shows a significant linear relationship

Notes: Each point is a country-year; r, p, n per panel. Nigeria 2012 (+172%) is a FAOSTAT reporting artifact.

Source: Author computations; see Table 3 notes.

Frictions matter at the margin. EU pesticide (ethylene oxide) controls forced a documented origin diversification away from India in 2021–22 (Observed, B). NCDEX sesame futures exist but are illiquid: no effective hedge (Observed, B). China's ∼379,000 tonnes of port stocks cover about four months of imports (Observed, B). The FASTER Act had a perverse demand effect: bakers added sesame to recipes, making demand stickier (Observed, B).

6.4 Downstream: the price channel is irrelevant; the operational channel binds

Sesame per seeded bun is 1.68 g on the single disclosed industry spec sheet (Låntmannen Unibake, January 2026: 2% on an 84 g bun; Observed, B). At the $1.60/kg food-grade benchmark (Commodity Board, 2025), that is $0.0027 per bun, about 0.27 cents (Reconstructed, B). Figure 10 puts it in context: 0.90% of an estimated $0.30 contract bun, 0.15% of sandwich food cost, 0.016% of restaurant annual food cost, and 0.022% of EBITDA after a 50% price spike.

A 50% sesame price spike costs the average US McDonald's restaurant about $98 per year and the whole US chain about $1.3 million (Modelled, B). Even a 500% spike, to $9.60/kg, costs $981 per restaurant per year, 0.2% of EBITDA (Figure 11). The three chains combined use roughly 6,750 tonnes a year, about 0.1% of world production (Modelled, C): burger buns are a rounding error in the sesame market.

The vulnerability is operational, not financial, and it has three strands. First, the approved-supplier

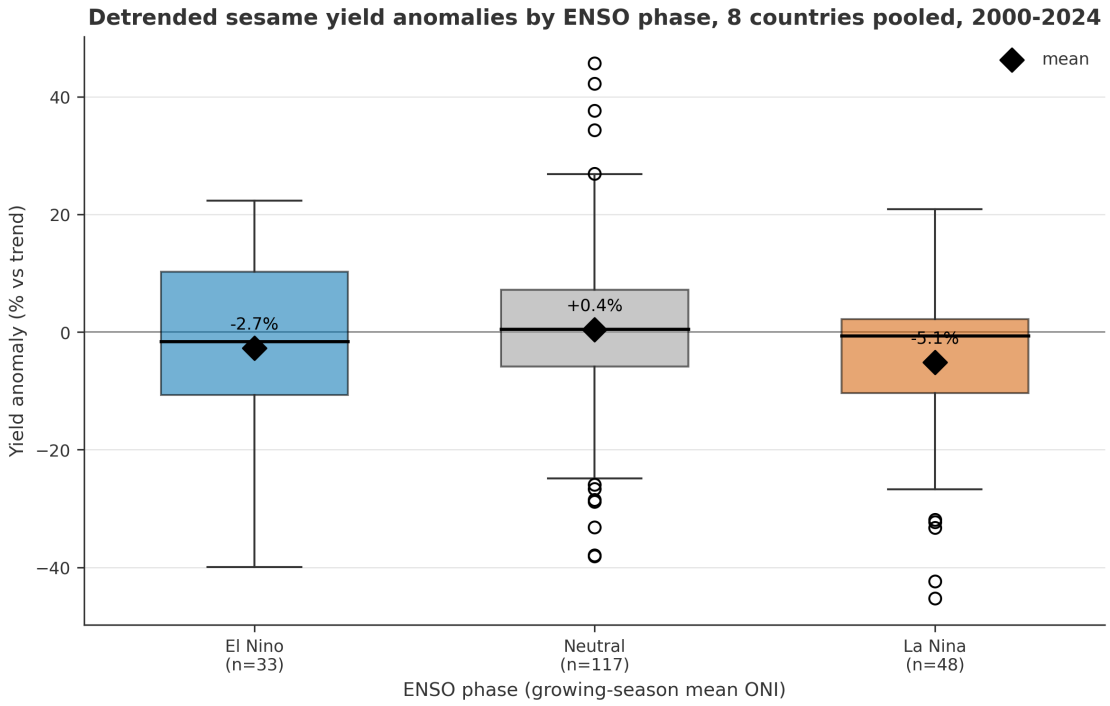


Figure 5: Yield anomalies by ENSO phase, pooled across eight countries
Notes: ONI averaged over each country's growing season; ≥+0.5 El Niño, ≤-0.5 La Niña. Two documented FAOSTAT reporting breaks excluded.
Source: Author computations from FAOSTAT and NOAA CPC.

system is rigid: franchisees must buy from named bakeries, and qualifying an alternate sesame origin or supplier takes an estimated 2 to 6 weeks of spec verification, lab testing, allergen audit, and trial bakes (Assumed, C; industry practice, not chain-disclosed). A 2-to-4-week shortage of food-grade sesame can therefore outlast the re-approval window. Second, the FASTER Act reformulations removed the plain-bun fallback: where sesame flour is now in the formula (Wendy's buns and French toast sticks, Chick-fil-A buns, Olive Garden breadsticks), dropping the topping does not remove the allergen and does not solve a shortage (Observed, B). Reformulating to plain buns requires corporate spec approval, new packaging, and re-labeling across the distribution network, a matter of weeks. Third, allergen recalls are the documented failure mode: three sesame recalls since 2022, two Class I (Aunt Millie's buns, September 2024; Prime Foods steamed buns, August 2024), plus a 2023 recall after a child suffered a reaction (Observed, A). Recall, liability, and brand costs at the flagship product dwarf any ingredient-price scenario.

No documented case was found of a sesame shortage interrupting quick-service bun supply specifically; the interruption scenario is reasoned from the supplier system's structure, not from precedent (gap acknowledged).

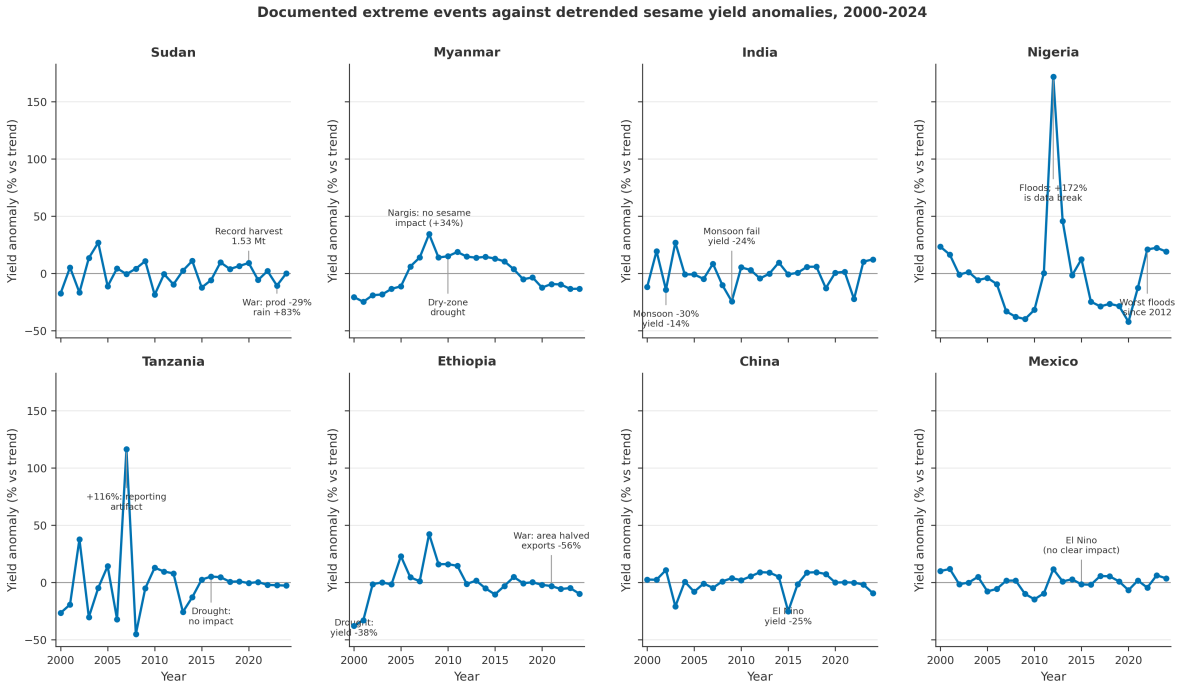


Figure 6: Sesame yield timelines with annotated extreme events, 2000–2024
Notes: See text for episode documentation and attribution.
Source: Author computations from FAOSTAT and weather sources above.

7 Scenarios, 2027–2035

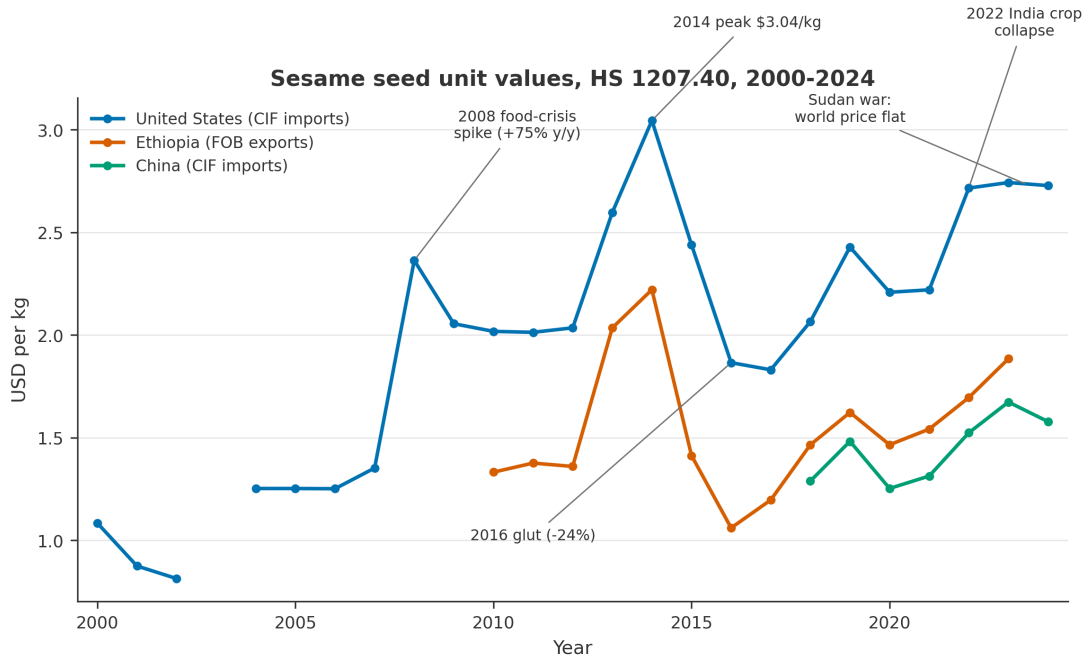
Five scenarios over nine years, with subjective probabilities calibrated to 2000–2024 historical frequencies (all Modelled, C). Figure 12 shows the price paths.

S1. Baseline: orderly growth (P ∼55%). Production grows 2–3%/yr on new-origin expansion; regional shocks occur at the historical rate of 1–2 per decade but substitution absorbs them, per the Sudan 2023–24 precedent. Prices mean-revert in a $1.50–1.80/kg band. No material consequence for restaurant chains.

S2. Recurring ENSO disruption (P ∼30%). Two or more significant La Niña events; each drags world output 5–12% and lifts prices 8–18% for a single crop year. Brief margin squeeze for fixed-price bakery contracts at rollover; inventory holders gain windfalls. Restaurant cost impact stays trivial.

S3. Accelerating warming drag (P ∼40%). Yield drag of 0.5–1.0%/yr in the most exposed regions (Sahel belt, lowland India) as flowering windows exceed thermal limits more often and charcoal-rot pressure rises. Suitability modeling for China finds contraction of suitable area exceeding expansion under warming scenarios (MDPI Agronomy, 2024). World production growth slows toward 1%/yr; real prices drift 2–3%/yr, cumulating to 20–30% elevation over nine years. No crisis, but a structural cost ratchet and a sourcing shift toward higher-latitude or irrigated origins.

S4. Severe multi-region failure (P ∼12%). A compound trigger, strong La Niña plus positive



Source: UN Comtrade public preview API, HS 120740, retrieved 2026-10-07/08. US 2003 missing (fetch failure); US 2000-2005 Comtrade-estimated.

Figure 7: US sesame import unit value (CIF, $/kg), 2000–2024, with shocks annotated
Notes: 2003 missing (Comtrade fetch failure); the line breaks there and nothing is interpolated. Pre-2006 values are Comtrade-estimated.
Source: UN Comtrade HS 1207.40, retrieved 2026-10-08.

IOD plus a regional conflict or export restriction, cuts world supply 20–30%. Prices rise 31–46% centrally, up to 85% with the export-share multiplier, for 1–2 crop years. China's port stocks blunt the first four months. Processors on fixed-price bakery contracts face a margin crisis; restaurant chains face the operational risk (re-approval, re-labeling, the sesame-bun brand decision) rather than a cost crisis.

S5. Structural deficit (P ∼10%). Demand outruns supply persistently: China import growth, FASTER Act demand stickiness, Sudan non-recovery, continued Chinese production decline. Sustained 50–100% price elevation over 5–9 years. Bakery contracts reprice structurally; integrated processors with established supply chains gain pricing power; quick-service procurement teams manage a seller's market under allergen rules.

The joint probability of at least one material disruption (S2 or S4) in the window is roughly 35–40%. The null weather-yield result cuts both ways: climate is a weak predictor of normal-year outcomes, but conflict and policy dominate the observed shocks, so S4's trigger is more likely geopolitical than meteorological. Leading indicators: sustained ONI ≤ -1.0 (3–6 months lead), Sudan/Ethiopia conflict status, Qingdao stocks below 200 kt, Indian kharif planting down more than 15% year on year, ECX export volumes collapsing vs seasonal norms, and Indian export-policy signals.

Winners and losers: inventory holders and substitute origins gain in every disruption scenario;

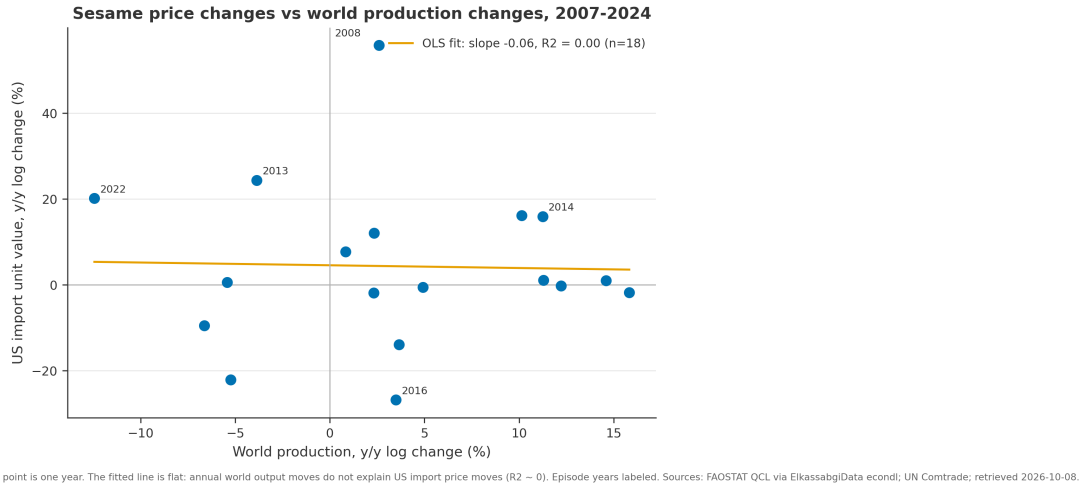


Figure 8: Annual US import price change vs annual world production change: no relationship ($R^2 = 0.00$)
Notes: $n = 18$ (2003 missing). Slope -0.06 (s.e. 0.59).
Source: Author computations from UN Comtrade and FAOSTAT.

shock-zone farmers lose (yield loss exceeds any price gain); fixed-price bakers get squeezed; EU buyers face MRL friction when forced onto new origins; integrated processors gain pricing power in structural tightness.

8 Limitations, robustness, and falsifiability

The FAOSTAT direct API was unreachable during research; production data come via a mirror, cross-checked on one published figure but not fully re-verified. Nigeria's and Tanzania's FAOSTAT series contain reporting breaks that make their correlations unreliable; pooled results are reported with and without Nigeria and remain null either way. Myanmar weather rests on ERA5 alone after POWER was rejected for that country; a third source is needed before any Myanmar weather-yield claim is publishable. The Indian Ocean Dipole is not in the yield panel. Sesame-belt weather points are judgment-based, not production-weighted, and national aggregation dilutes sub-national signals: a state-level panel could overturn the country-scale null without contradicting these numbers, and we state that as the primary falsifier.

The elasticity rests on a single episode (India 2022) and is an upper-magnitude bound. Grade-specific price premiums have no public series; processor and bakery inventory levels are unknown; no chain discloses sesame volumes, per-bun content, or bun contract prices, so all downstream figures downstream of those inputs are modeled estimates with stated ranges. No quick-service sesame shortage precedent was found, so the interruption scenario is structural reasoning, not empirical.

Falsifiers: a sub-national panel showing $|r| > 0.4$ for rainfall would overturn the country-scale null; a $> 15\%$ world output cut with $< 10\%$ price response would break the elasticity band; chain-disclosed

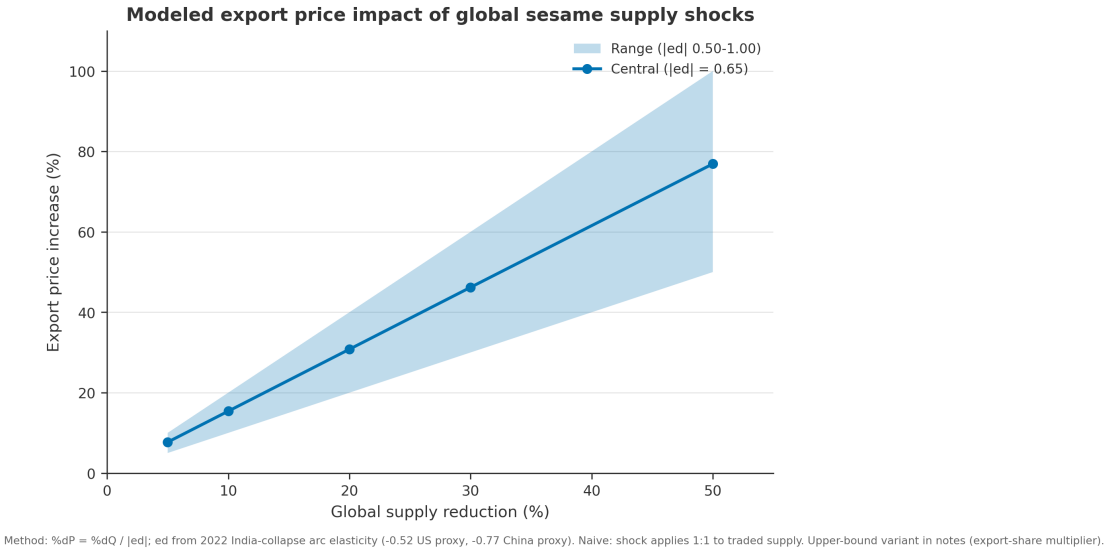


Figure 9: Modeled export price impact of global sesame supply shocks
Notes: Central elasticity −0.65, range −0.50 to −1.00, from the 2022 India episode. Assumes the shock applies 1:1 to traded supply.
Source: Author computations.

procurement costs showing material margin impact would overturn the downstream arithmetic; La Niña events in 2027–2030 with no yield drag would mark the −5.1% as noise. This is a retrospective analysis; there was no pre-registered plan.

9 Conclusion

Three claims, each hedged with its evidence grade. First, the sesame supply chain is concentrated in rainfed smallholder systems with no inventories beyond commercial pipeline stocks, and its largest recent shock was war, not weather (Observed, A). Second, at the scales that matter for prices, neither weather nor production moves the market much: country weather does not explain country yields (Reconstructed, B), and world production does not explain world prices (Reconstructed, B), because substitution across origins absorbs most shocks. Third, the economic exposure that does exist sits at the end of the chain and is operational, not financial: allergen law has locked sesame into bun formulas inside a rigid approved-supplier system, and the documented failure mode is recalls and re-approval delays, not ingredient cost (Observed, B).

The practical reading: monitor the leading indicators in Section 7, not the price. A sesame crisis that matters to a restaurant chain will arrive as a shortage of qualifying seed, not as a price spike. The 2027–2035 scenarios put 55% weight on orderly absorption, 35–40% on at least one material disruption, and 10% on sustained structural deficit. None of them makes sesame expensive enough to matter to a P&L. All of the disruptive ones make it hard enough to source to matter to operations.

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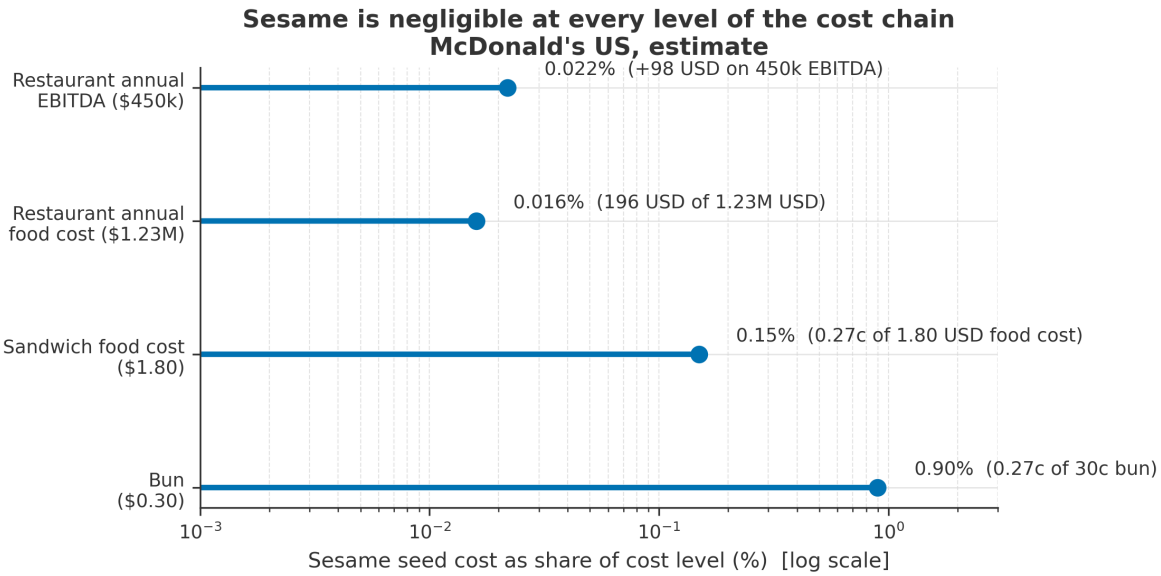


Figure 10: Sesame is negligible at every level of the cost chain (McDonald's US, estimate)
Notes: Log scale. Bun contract price ($0.30), sandwich food cost ($1.80), and franchisee EBITDA ($450k) are modeled estimates; no chain discloses them.
Source: Author computations; see text.

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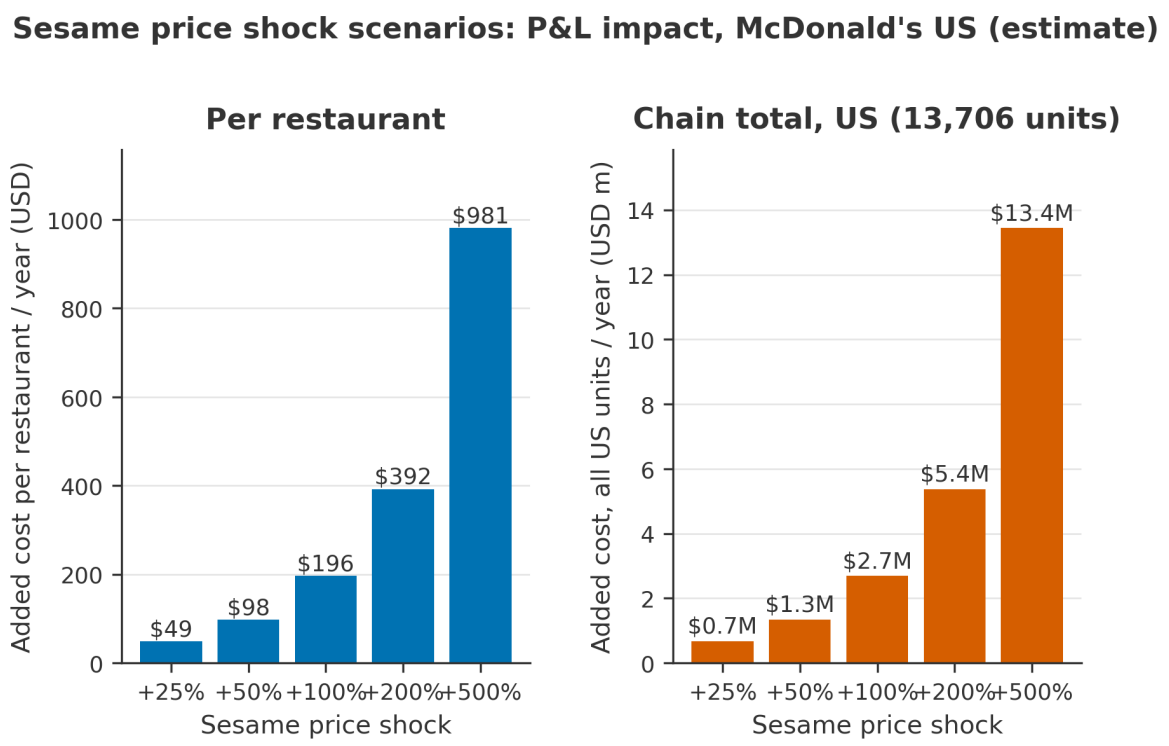


Figure 11: Annual cost of sesame price shocks, per US McDonald's restaurant and chain-wide
Notes: Estimates; see cost-share derivations in text.
Source: Author computations.

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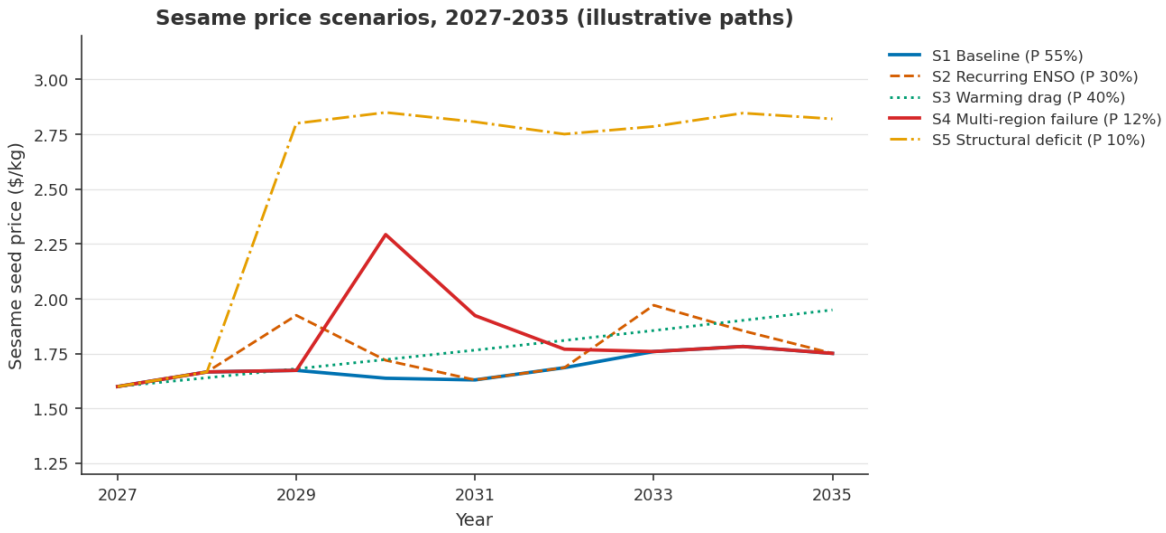


Figure 12: Sesame price paths under five scenarios, 2027–2035
Notes: Base $1.60/kg. Probabilities are subjective judgments; S2 and S3 overlap.
Source: Author computations.

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A Data appendix

Table 4 defines the series. All computations were performed in Python; scripts and CSVs are archived in the paper's data directory.

Table 4: Series definitions and sources

Series	Definition	Source
Production	Tonnes, sesame seed, item 289	FAOSTAT QCL via mirror
Yield	hg/ha, production / area	FAOSTAT QCL via mirror
Yield anomaly	% deviation from country OLS trend	Author computation
Rain anomaly	% deviation, growing season vs 1991–2020	NASA POWER / ERA5
ONI phase	Growing-season mean ONI $\geq$+0.5 / $\leq$-0.5	NOAA CPC
US import price	Unit value, CIF, $/kg, HS 1207.40	UN Comtrade
Trade flows	Tonnes and USD, HS 120740	UN Comtrade
Bun sesame	1.68 g/seeded bun	Låntmannen Unibake spec, Jan 2026

Notes: Sudan splices pre-2011 borders; South Sudan separate from 2012. Ethiopia splices PDR series. China = mainland only.
Source: As listed.